

Fault Distance Localization Method for Heterogeneous Distribution Networks

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Abstract—In many cases the capacity of a network to endure a fault without ensuing major problems or long blackouts is related to the capacity to locate and isolate the faulty section. Transmission grids already have in general an evolved fault localization mechanism thanks to many distance protections, unlike distribution grids. This paper describes a fault localization method for heterogeneous distribution networks that could be deployed thanks to today's smart grid developments. The present method is valid for networks with a neutral grounding through an impedance and for single or multiphase near bolted fault.

Index Terms— distribution network, fault distance, fault localization, heterogeneous network, Takagi

I. INTRODUCTION

Fault localization is an important function in electrical networks. It is known that if a fault is not isolated fast enough, major problems can follow [1]. But a fault not located fast enough, or wrongly located, can also ensue problems since its precise location is needed in order to correctly reconfigure the network. Quickly and precisely locating a fault is therefore an important feature. The transmission grid operation is much more critical in case of a fault, consequently fault localization is quite developed. Hopefully the transmission grid is rather simple and the developed methods are working well.

The distribution network however is generally not as well monitored as the transmission network. And its complexity due to its radial structure, or its strong heterogeneity, is such that today real time fault localization is either not developed or very basic. The French localization method for example consists, in general, in isolating the fault, then using fault indicators to find a faulty section, before doing a final precise localization. These steps are done once the faulty section is isolated from the rest of the network.

Considering today's development of smart grids [2], it is possible to review these techniques and propose new fault localization methods. Smart grid developments will generalize the monitoring of feeders, introducing new measurements, not previously accessible, but also precision and better knowledge

of network's topologies. The developed methods can therefore count on numerous and precise data for each feeders of the distribution network.

II. USING TAKAGI'S LOCALIZATION ALGORITHM

Various localization techniques can be used to locate a fault: positive sequence reactance [3], loop reactance method [4] and the Takagi method [5]–[6] are the main loop impedance based methods. The Takagi algorithm which has been chosen here was initially developed for the transmission grid, but can be adapted to the distribution network.

The initial Takagi formula can be obtained from the symmetric components equations of the following network for a resistive single line to ground fault:

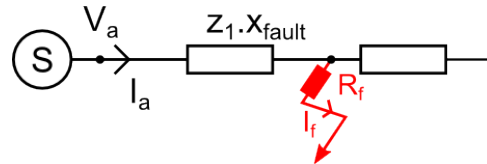


Figure 1. Simple network with a resistive single line to ground fault

Summing and rearranging the equations we find:

$$V_a = (z_1 \cdot x_{fault}) \cdot (I_a + K \cdot I_0) + R_f \cdot I_f \quad (1)$$

V_a	Voltage of phase a
z_1	Per unit length positive line impedance
x_{fault}	Fault distance
I_a	Current in phase a
K	$= \frac{z_0 - z_1}{z_1}$, constant
I_0	Zero sequence current
R_f	Fault resistance
I_f	Fault current

In (1) the terms R_f and I_f are unknown. Removing these terms is possible by multiplying both sides of the equation by I_{ideal}^* , with I_{ideal} an ideal current in phase with I_f , and taking only the imaginary part of the resulting equation. Isolating x_{fault} we then find:

This work is part of the Greenlys project, supported by the ADEME (Convention n°93PF92001D) throughout the smart grid program
Programme Réseaux Electriques Intelligents des Investissements d'Avenir

$$x_{fault} = \frac{\Im(V_a \cdot I_{ideal}^*)}{\Im(z_1 \cdot (I_a + K \cdot I_0) \cdot I_{ideal}^*)} \quad (2)$$

A. Inputs and Parameters

The developed equation (2) is quite simple to use since it uses a small number of parameters and measures. Considering a homogeneous network (all lines are identical), the only parameters to know are the per unit length direct and zero sequence impedances of the line, z_1 and z_0 . The only needed measures are the faulted phase voltage and the three phase currents.

B. Approximations

In this approach two main approximations are made up to this point. One related to the current flowing through the line elements and one related to the chosen current in phase with I_f :

- We consider in this network that the total current I_a flows in the line element impedance Z_1 , and that the currents drawn by possible loads located between the measurement and fault points are neglected. We will develop in III a method that overcomes this approximation.
- The second approximation is to consider that the chosen current I_{ideal} is perfectly in phase with the fault current so that the term with the fault resistance R_f has no impact in the calculations.

III. MANAGING LOCALIZATION ON A HETEROGENEOUS NETWORK

The localization formula described in II seems to be easy to apply. The needed measurements and parameters are not too numerous, and the complexity of the calculation is not high. However, apart from the two approximations listed above, the main problem of the method is that it needs a homogeneous network. Since most distribution networks are clearly heterogeneous, we will develop hereafter a method dedicated to heterogeneous networks.

A. Problems induced by heterogeneity

With a heterogeneous network, the parameters z_1 and z_0 in (1) and (2) cannot be known precisely. A lesser evil is to determine a method for choosing an apparent per unit length impedance. For example, calculating a mean impedance for the whole network, only for the main line, or using the impedance of the most common line type of the network, etc. But the effect will be to bring more approximations and bad results.

One of the consequence of the smart grid developments occurring today will be to have access to numerous precise voltage/current measurements on the networks. Phase voltages and currents at the head of a feeder should soon always be available. Another consequence is the increase of automated operations that will be done by the Distribution Network Operators (DNOs). New software solutions will be developed. Consequently, we can count on a full digital topology description of the networks.

A distribution network is described as nodes connected by line elements. The needed information about the network to develop localization methods consists of the length and the electrical characteristics of each line elements, and the way nodes are connected by the lines. Active and reactive powers of the load located on the network will also be needed.

B. Method for localization on heterogeneous networks

As stated previously, the developed localization method in II works for a homogeneous network. However a heterogeneous network is only a sequence of homogeneous sections. The idea is therefore to place the algorithm upstream of a section and make a localization on the section's length only. If the found fault distance is bigger than the section's length, the calculation needs to be done for the next downstream section.

In order to have an efficient fault localization algorithm, two modules need to be developed: a systematic browsing of the tree structure of the network has to be done and the voltage and current measurements at the head of the feeder need to be spread at the head of each homogeneous section.

The principle of the method is based on the quadripole theory. Each network element can be considered as an eight-pole element (4 inputs/4 outputs for the 3 phases and the neutral) [7]. Converting the "phase" representation of the element to a "symmetrical components" representation gives three independent four-pole elements:



Figure 2. Four-pole element

Two 2×2 matrices link the voltage/current input and voltage/current output of the quadripole:

$$\begin{bmatrix} V_{out} \\ I_{out} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \cdot \begin{bmatrix} V_{in} \\ I_{in} \end{bmatrix} = [B] \cdot \begin{bmatrix} V_{in} \\ I_{in} \end{bmatrix} \quad (3)$$

The output voltages and currents of the four-pole element can be computed from the input voltages and currents thanks to the B matrix.

In order to compute the fault distance on a homogeneous section, we therefore need to know the four b parameters which link the substations measurements V_{in} and I_{in} to V_{out} and I_{out} , the measurements at the head of the homogeneous section. The final B_{eq} matrix will be the product of series B matrices corresponding to line and load elements located between the measurement point and the propagation point.

Therefore the first step, which can be done once for a given network topology, is to compute beforehand each B matrix linking two consecutive nodes. From these matrices, every B_{eq} matrix linking any propagation point with the measurement point can be computed. To compute the equivalent B matrices, each component in the network has to be described with a B type matrix. The components that can be found between two nodes are a line or a load (or loads

aggregated in a single load). We can also find a branch (bifurcation) but it can be aggregated to a “load”. Therefore we have the following components and quadripoles:

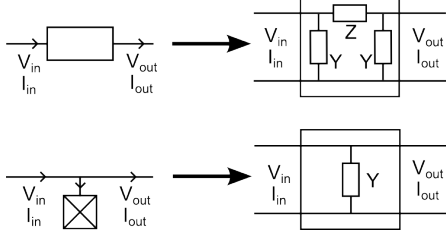


Figure 3. Quadripoles representing a line (top) and a load (bottom)

The associated B matrices for these two quadripoles are detailed in TABLE I.

TABLE I. B MATRICES FOR LOAD AND LINE ELEMENTS

	[B]
Load	$\begin{bmatrix} 1 & 0 \\ -Y & 1 \end{bmatrix}$
Line	$\begin{bmatrix} 1+Y.Z & -Z \\ -2.Y-Y^2.Z & 1+Y.Z \end{bmatrix}$

Having computed each B matrix linking two consecutive nodes it is now possible to compute the voltage and current in any point of the network from the measurements made at the head of the feeder.

As stated earlier, the aim of this tool is to manage the heterogeneity of the network which prohibits from computing a fault distance directly from the measurement point. Considering the heterogeneous network as a set of homogeneous sections and being able to obtain the voltage and current at the head of each homogenous section solves this problem.

Furthermore this method has another advantage: the approximation stated in II.B on the current drawn by possible loads between the measurement and fault points is not an approximation anymore. Indeed the algorithm computes (2) on a homogeneous section which is composed of a single line element.

IV. APPLICATION ON A FRENCH RURAL NETWORK

In the following we will present fault localization results obtained by the application of this method to a French rural 20 kV feeder simulated in EMTP/ATPDraw.

The data concerning the feeder gives details on the nodes, loads and line sections' connections. We had access to the real electrical characteristics of each section as well as their length and the active and reactive peak power of the loads.

A. The Considered Network

The network is a feeder with a peak load of 3095 kVA and a capacitive current of 15A. The feeder is of radial type and composed of a main feeder of 21.507 km, three long laterals between 2.545 km and 6.933 km and many other laterals from a few hundred meters to one or two km. It corresponds to 270 different line sections, from a few meters to one kilometer. Seven different types of line compose the feeder, with the

following distribution (TABLE II.). This network is therefore interesting by its strong heterogeneity and radial nature.

In EMTP/ATPDraw the line/cable model takes into account propagation and transposition. This model differs slightly from the PI model we use (III.B) for the propagation of voltage and current.

TABLE II. CONDUCTOR'S TYPE AND LENGTH IN THE NETWORK

Type (mm ²)	Line (148)	Line (76)	Line (54)	Line (34)	Cable (95)	Cable (150)	Cable (50)
Number	14	9	117	101	25	3	1
Length (m)	11343	3245	28354	28234	4270	447	58

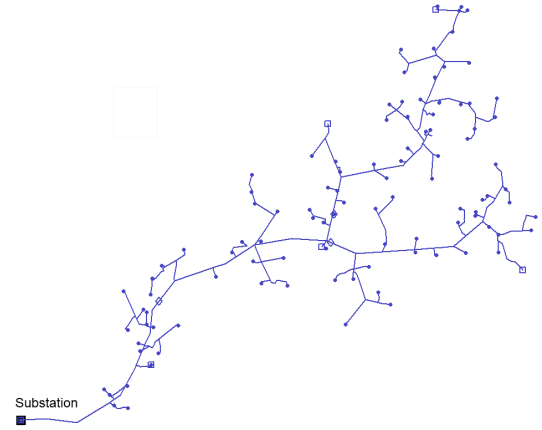


Figure 4. Rural network considered in the study

B. Results

1) Voltage and current propagation precision

The fault localization computation is based on the voltage and current propagated at the head of the homogeneous line section considered. Therefore the precision of this propagation step is major in order to make a precise fault localization.

In normal operation condition, we obtain an overall voltage error (relative difference between propagated values and simulation measurements) always inferior to 0.013%. Concerning the current, as it decreases along the feeder, the relative error increases when propagating near the end of the feeder. However, up to 17km the current error is always less than 1%. These results validate the propagation of the measured voltage and current through the feeder. The precision in fault condition is specific to each fault and depends of too many parameters to be able to give a general number. However, the good localization results which follow show that the precision is good enough even in fault condition.

2) Fault localization results

Thirty one different fault locations have been simulated on this network. Nineteen faults are located on the main feeder and twelve faults are located on laterals of the main feeder. The fault type is a bolted phase-to-ground fault (fault resistance of 0.1mΩ).

When a fault is detected, the algorithm will be provided with the time domain values of the voltage and current measured at the head of the feeder during 0.02s, starting 0.01s

after fault occurrence when the fault transient is still not over. The algorithm will then browse through the feeder, propagating the measured voltage and current, and performing a fault location calculation for each homogeneous section. When the algorithm finds a possible fault on a section, it does not browse further downstream, but browses into possible upstream bifurcation if there are any. Consequently the algorithm will find one or more possible fault location due to the radial structure of the feeder. The found fault location will not necessarily be at the same distance of the measurement point because of the heterogeneity of the feeder.

The precisions below are given in order to help understand the results which will follow:

- For each fault simulation the algorithm will give one or more section found as “possibly faulty”
- For each section found “possibly faulty” the algorithm gives an associated fault distance
- Considering that for the ease of simulation faults have been placed at the junction of homogeneous sections, for each fault the algorithm can find either the unique upstream section, or the downstream section(s) connected to the junction point

We therefore consider a correct section localization only if among the proposed section, the algorithm gives the upstream section or all of the downstream sections connected to the fault point.

a) Distance results

We will first analyze the results in terms of the found fault distance compared to the real fault distance for each fault location. The general performance (Figure 5.) of this algorithm on this study case is a maximum distance error of 1.6% (fault number 7; distance error of 129m). The absolute maximum distance error being 163m (fault number 10).

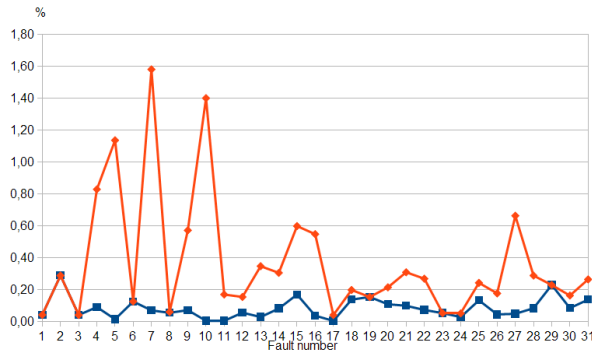


Figure 5. Distances error in % of fault distance for each fault location. For the distances associated to the correct results only (blue), to all results (red)

It is important to understand that the algorithm will find one or several fault location, each associated with a couple distance/total equivalent impedance, which will vary depending if the found fault location is on a highly loaded branch, or lowly loaded lateral for example. It is therefore inaccurate to talk about a distance error for found fault location not corresponding to the real fault location. Hence, considering only the distances errors of the found faulty

sections correctly corresponding to the actual fault location, the maximum distance error is of 0,29% (fault number 2; distance error of 8m), and the absolute maximum distance error is 34m (fault number 29).

b) Section localization

As stated previously, for each fault location the algorithm will give one or more “possibly faulty” section. The first important result is that for the thirty one fault cases, the algorithm always finds (among others) either the unique upstream either all the downstream sections connected to the junction point where the fault is located.

Due to the radial structure of the feeder, the algorithm gives most of the time other answers than only the correct faulty sections. Indeed for a given fault, it is highly probable that there are one or more other fault locations compatible with the voltage/current couple measured at the head of the feeder. TABLE III. below shows the occurrences of the number of proposed “possibly faulty” sections for the 31 fault simulated.

TABLE III. OCCURRENCES OF THE NUMBER OF PROPOSED SECTIONS

Nb. of proposed sections	1	2	3	4	5	6	7	8	9	12
Occurrences	4	3	6	4	4	4	3	1	1	1

For this network, the maximum number of proposed sections is 12 and occurred only once. There are 3 to 6 occurrences for a number of proposed sections between 1 and 7. These numbers will of course be dependent of the structure of the feeder, the length and number of its laterals.

An interesting analysis is to look at the mean ratio of correct faulty section answers depending of the total number of proposed sections. We can see in Figure 6. that when the algorithm proposed only one possibly faulty section (case which occurred 4 times, TABLE III.), it was always the correct faulty section.

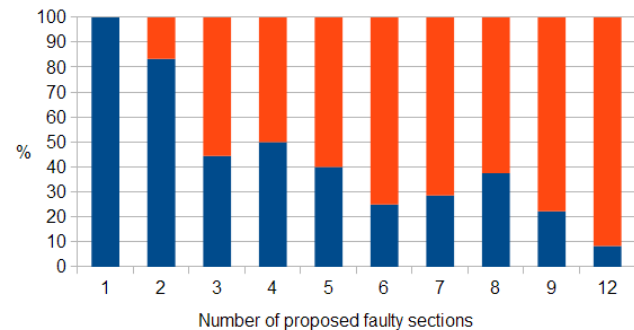


Figure 6. Mean percentage of right (blue/dark) and wrong (red/light) answers depending of the number of proposed sections

The mean correct ratio for solutions with 2 proposed section is of 83%. This ratio quickly falls below 50% and decreases logically as the number of proposed solutions increases. These percentages cannot be generalized to every situation. First of all they are not representative enough (only 31 fault location simulated) and they will depend on the

structure of the network as well as where the faults are chosen to be simulated. But these results confirm an important point: this localization algorithm needs to be associated with other methods in order to discriminate between all the proposed sections which are the ones that can really be faulty. Therefore the performance of this localization algorithm cannot be judged only by its distance performance, or correct faulty section proposition, but also by the completeness of its results.

In the following the completeness of the algorithm is shown for one of the thirty one simulated fault.

c) *Example with fault number 13*

This fault, located at 14585m from the measurement point, is on a junction with 3 downstream sections. The algorithm effectively finds the 3 downstream sections as well as 5 other sections on laterals (Figure 7.).

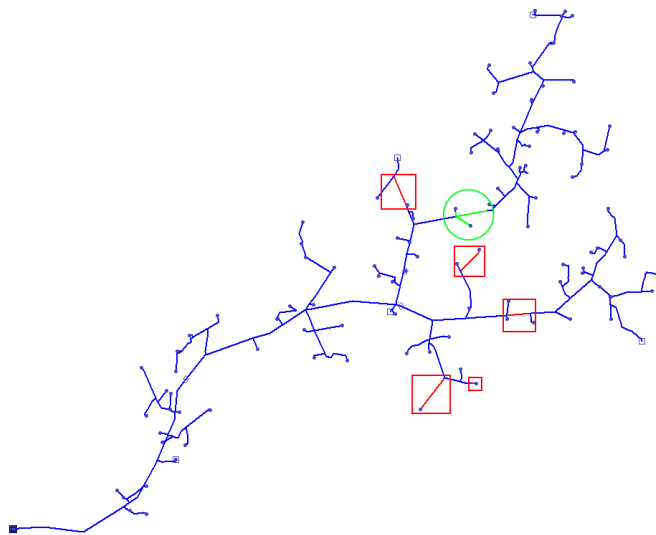


Figure 7. Display of the network with the 3 correct faulty sections (green/circle) and the 5 other proposed sections by the algorithm (red/square)

TABLE IV. shows for each found section the distance, localization status, distance errors in meters and relative distance errors for the correctly faulty sections found by the algorithm for the fault number 13.

TABLE IV. LOCALIZATION RESULTS FOR FAULT NUMBER 13

Status	Section n°	Distance found (m)	Dist. Error (m)	Dist. Error (%)
correct	117	14589,2	4.2	0.029
correct	168	14589,2	4.2	0.029
correct	172	14588,6	3.6	0.025
wrong	37	14635,8	NR ^a	NR
wrong	19711	14635,0	NR	NR
wrong	5	14598,6	NR	NR
wrong	63	14624,3	NR	NR
wrong	59	14624,0	NR	NR

a. NR: Not Relevant – see IV.B.2)a for further details

As explained previously, there is no “distance error” for a possible fault location which does not corresponds to the real

one. The fault number 3 is located at the junction of sections 117, 168 and 172 at a distance of 14585m. The algorithm finds all the other possible faulty sections which have a slightly different total impedance and therefore distance.

Again, it is necessary to discriminate the results of the algorithm, which are sections associated to a fault distance, in order to find the real faulty section. Using smart grid attributes such as voltage/current measurement units along the feeder could be a first step in fault location discrimination. Knowing the voltage in the middle of the feeder during the fault divides by 2 the possible faulty zone.

V. CONCLUSIONS

As it has been shown in IV.B.2 the present algorithm proves to be quite effective for heterogeneous distribution networks. By the relative small absolute distance errors compared to the network’s size, by its exhaustive possibly faulty section propositions and by its ease to run with few real time data this method is an interesting step in fault localization for distribution networks. The next steps are to make the algorithm robust to various situations. Arc type faults will be looked into for example. It will also be interesting to develop the algorithm for compensated neutral grounding network. Other tests need to be carried in order to evaluate the influences of resistive faults, of the loading of the feeder on fault occurrence, or the capacitive currents of other healthy feeder connected to the substation.

Finally we hope to be able, through the Greenlys project’s collaborations, to confront this algorithm to real fault measurements. It is quite certain that the algorithm’s performance will decrease and it will be interesting to develop the method for a real practical application.

ACKNOWLEDGMENT

The authors would like to thank *ERDF Bourgogne* whom kindly accepted to provide us with real data in order to produce this work.

REFERENCES

- [1] D. Koval & A. Chowdhury, "Reliability Analysis of the Isolation and Restoration Procedures of Distribution Feeders," in *Proc. 2006 IEEE PES Transmission and Distribution Conference and Exhibition*, pp. 766-773.
- [2] G. Taylor; D. Wallom; S. Grenard; A. Yunta Huete & C. Axon, "Recent developments towards novel high performance computing and communications solutions for smart distribution network operation," in *Proc. 2011 2nd IEEE PES International Conference and Exhibition on Innovative Smart Grid Technologies Europe*, pp. 1-8.
- [3] S. Kulkarni; N. Karnik; S. Das & S. Santoso, "Fault Location Using Impedance-Based Algorithms on Non-Homogeneous Feeders," in *Proc. 2011 IEEE Power and Energy Society General Meeting*
- [4] N. Karnik; S. Das; S. Kulkarni & S. Santoso, "Effect of Load Current on Fault Location Estimates of Impedance-based Methods," in *Proc. 2011 IEEE Power and Energy Society General Meeting*
- [5] K. Zimmerman & D. Costello, "Impedance-Based Fault Location Experience," in *Proc. 2006 IEEE Rural Electric Power Conference*, pp. 1-16.
- [6] T. Adu, "A New Transmission Line Fault Locating System," *IEEE Trans. Power Delivery*, vol. 16, pp. 498-503, 2001.
- [7] T. Welfonder; V. Leitloff; R. Fenillet & S. Vitet, "Location strategies and evaluation of detection algorithms for earth faults in compensated MV distribution systems," *IEEE Trans. on Power Delivery*, vol. 15, pp. 1121-1128, 2000.